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BMATE101 – MATHEMATICS-I (EEE)

MODULE-1 : CALCULUS

POLAR COORDINATES & POLAR CURVES

1. Polar Coordinates

In polar coordinates, the position of a point is defined by:

- $r \rightarrow$ distance from origin (pole)
- $\theta \rightarrow$ angle made with the initial line (polar axis)

A point is represented as:

$$(r, \theta)$$

Relation between Cartesian and Polar coordinates:

$$x = r \cos \theta, y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}, \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

2. Polar Curves

A **polar curve** is a curve represented by an equation of the form:

$$r = f(\theta)$$

Common Polar Curves

Curve	Equation
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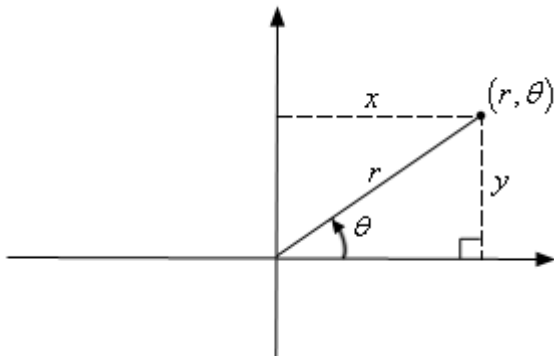
Circle	$r = a$
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Straight line	$r \cos \theta = a$
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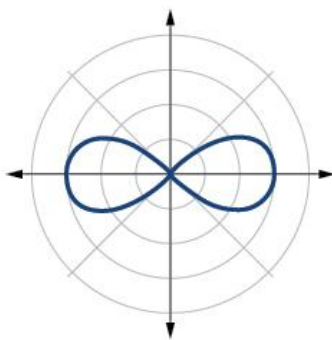
Cardioid	$r = a(1 + \cos \theta)$
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Rose curve	$r = a \cos n\theta$
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Polar Curve Diagram (REFERENCE)



Lemniscate



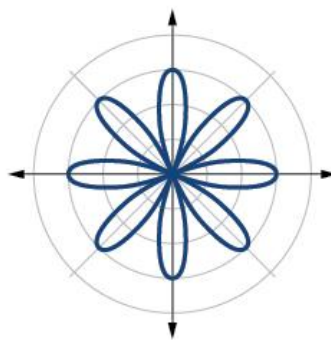
$$r^2 = a^2 \cos 2\theta$$

$$r^2 = a^2 \sin 2\theta$$

$$a \neq 0$$

(a)

Rose Curve (n even)



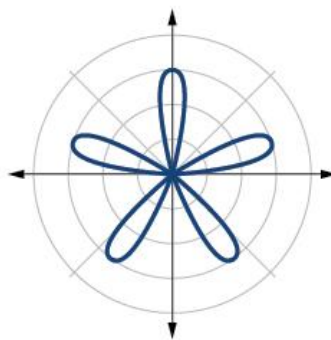
$$r = a \cos n\theta$$

$$r = a \sin n\theta$$

$$n \text{ even, } 2n \text{ petals}$$

(b)

Rose Curve (n odd)



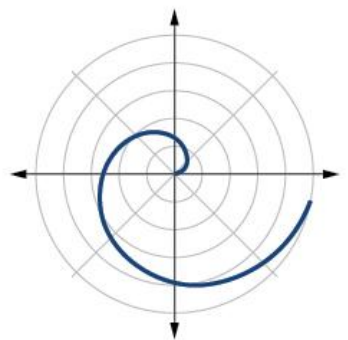
$$r = a \cos n\theta$$

$$r = a \sin n\theta$$

$$n \text{ odd, } n \text{ petals}$$

(c)

Archimedes' Spiral



$$r = \theta$$

$$\theta \geq 0$$

(d)

3. Angle Between Radius Vector and Tangent

Let the polar curve be:

$$r = f(\theta)$$

The angle ψ between the **radius vector** and the **tangent** is given by:

$$\tan \psi = \frac{r}{\frac{dr}{d\theta}}$$

This formula is used to find:

- Direction of the tangent
- Orthogonality of curves

4. Angle Between Two Polar Curves

If two curves intersect at a point and their angles with the radius vector are:

$$\psi_1, \psi_2$$

Then the angle between the curves is:

$$|\psi_1 - \psi_2|$$

5. Pedal Equation

Definition

The **pedal equation** of a curve expresses the relation between:

- $r \rightarrow$ radius vector
- $p \rightarrow$ length of perpendicular from pole to the tangent

Pedal Equation Formula

$$p = r \sin \psi$$

where ψ is the angle between radius vector and tangent.

6. Curvature

Definition

Curvature measures **how sharply a curve bends** at a given point.

- Higher curvature $\rightarrow$ sharper bend
- Lower curvature $\rightarrow$ flatter curve

7. Radius of Curvature

The **radius of curvature** ρ is the reciprocal of curvature.

$$\rho = \frac{1}{\text{curvature}}$$

8. Radius of Curvature – Cartesian Form

For a curve:

$$y = f(x)$$
$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left|\frac{d^2y}{dx^2}\right|}$$

9. Radius of Curvature – Parametric Form

If:

$$x = f(t), y = g(t)$$

$$\rho = \frac{\left[\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right]^{3/2}}{\left| \frac{dx}{dt} \frac{d^2y}{dt^2} - \frac{dy}{dt} \frac{d^2x}{dt^2} \right|}$$

10. Radius of Curvature – Polar Form

For a polar curve $r = f(\theta)$:

$$\rho = \frac{\left[r^2 + \left(\frac{dr}{d\theta} \right)^2 \right]^{3/2}}{\left| r^2 + 2 \left(\frac{dr}{d\theta} \right)^2 - r \frac{d^2r}{d\theta^2} \right|}$$

11. Radius of Curvature – Pedal Form

If p is the length of perpendicular from pole to tangent:

$$\rho = r \frac{dr}{dp}$$

PROBLEMS ON CURVATURE & APPLICATIONS

1. Problems on Radius of Curvature (Cartesian Form)

Problem 1

Find the radius of curvature of the curve

$$y = x^2$$

at the point $x = 1$.

Solution

$$\frac{dy}{dx} = 2x \Rightarrow \frac{d^2y}{dx^2} = 2$$

Radius of curvature formula:

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}{\left| \frac{d^2y}{dx^2} \right|}$$

At $x = 1$:

$$\rho = \frac{(1+4)^{3/2}}{2} = \frac{5^{3/2}}{2} = \frac{5\sqrt{5}}{2}$$

2. Problems on Radius of Curvature (Parametric Form)

Problem 2

Find the radius of curvature for:

$$x = at, y = at^2$$

Solution

$$\frac{dx}{dt} = a, \frac{dy}{dt} = 2at$$
$$\frac{d^2x}{dt^2} = 0, \frac{d^2y}{dt^2} = 2a$$

Formula:

$$\rho = \frac{\left[\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right]^{3/2}}{\left| \frac{dx}{dt} \frac{d^2y}{dt^2} - \frac{dy}{dt} \frac{d^2x}{dt^2} \right|}$$
$$\rho = \frac{[a^2 + 4a^2t^2]^{3/2}}{|2a^2|}$$

3. Problems on Radius of Curvature (Polar Form)

Problem 3

Find the radius of curvature for:

$$r = a\theta$$

Solution

$$\frac{dr}{d\theta} = a, \frac{d^2r}{d\theta^2} = 0$$

Polar formula:

$$\rho = \frac{\left[r^2 + \left(\frac{dr}{d\theta} \right)^2 \right]^{3/2}}{\left| r^2 + 2 \left(\frac{dr}{d\theta} \right)^2 - r \frac{d^2r}{d\theta^2} \right|}$$

Substitute:

$$\rho = \frac{(a^2\theta^2 + a^2)^{3/2}}{a^2(\theta^2 + 2)}$$

4. Problems on Pedal Equation

Problem 4

For a curve, if the angle between the radius vector and tangent is ψ , show that:

$$p = r \sin \psi$$

Explanation

By definition:

- p is the perpendicular from pole to tangent
- Using triangle geometry at the point of contact:

$$p = r \sin \psi$$

This relation is used to convert equations into **pedal form**.

5. Applications of Curvature

Curvature and radius of curvature are used in:

- **Communication engineering** – antenna and waveguide design
- **Microphone and speaker manufacturing** – diaphragm shape optimization
- **Image processing** – edge detection and curve smoothing
- **Robotics** – trajectory planning
- **Mechanical design** – road and railway track curves

6. Important Results to Remember (Module-1)

- Angle between radius vector and tangent:

$$\tan \psi = \frac{r}{dr/d\theta}$$

- Radius of curvature (Cartesian):

$$\rho = \frac{(1+y'^2)^{3/2}}{|y''|}$$

- Radius of curvature (Polar):

$$\rho = \frac{(r^2 + r'^2)^{3/2}}{|r^2 + 2r'r'' - rr''|}$$

- Pedal relation:

$$p = r \sin \psi$$

7. Self-Study Topics (As per Syllabus)

(Only for reference, not usually asked in full-length problems)

- Centre of curvature
- Circle of curvature
- Evolutes
- Involute